

Face Recognition Using Eigenfaces and Pattern Classification

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DSCD612 Pattern Recognition, Project 3
MPhil/MSc Data Science, University of Ghana

Second Semester, 2025/2026

1 Problem and dataset

Closed-set face identification asks which of C enrolled individuals a given image depicts. Posed directly in pixel space the problem is badly conditioned: a 64×64 greyscale image is a point in $\mathbb{R}^{4096}$ and the dataset used here supplies 400 such points, so features outnumber observations by about ten to one. The sample covariance matrix is singular by construction and no per-class density is estimable.

Face images do not fill $\mathbb{R}^{4096}$. They concentrate near a much lower-dimensional structure, because faces share a common geometry. The Eigenfaces method [1] exploits this: it finds the linear subspace holding most of the variation across a set of faces, then recognises inside it. This report implements that pipeline from first principles, including the principal component analysis itself, to answer the project's central question:

How much dimensionality can be removed from facial images before identity-discriminating information is significantly lost?

The question admits two different measurements. *Reconstruction* asks how much pixel variance survives; *discrimination* asks how well identities can still be told apart. The main finding here is that the two criteria disagree by a factor of about four, for a reason that the component structure explains.

The **Olivetti (ORL) database** [2] supplies 400 greyscale images of 40 subjects, ten each, normalised to $[0, 1]$. Within a subject the images vary in lighting, expression, and the presence of glasses. All are cropped, frontal, and roughly aligned against a dark background, so this study measures the recognition method rather than a face *detection* front end. The redundancy that motivates the method is measurable: among 300 sampled pixels the mean absolute pairwise correlation is 0.26 and 10.4% of pairs exceed $|r| = 0.5$.

2 Method

2.1 Representation and principal component analysis

Each image is flattened row-major into $x_i \in \mathbb{R}^d$, $d = 4096$, and stacked into $X = [x_1, \dots, x_N]^T$ with $N = 400$. Writing $\mu = \frac{1}{N} \sum_i x_i$ and $A = X - \mathbf{1}\mu^T$, the sample covariance is

$$\Sigma = \frac{1}{N-1} \sum_{i=1}^N (x_i - \mu)(x_i - \mu)^T = \frac{1}{N-1} A^T A. \quad (1)$$

PCA maximises the projected variance $v^T \Sigma v$ subject to $v^T v = 1$. With a Lagrange multiplier, $\mathcal{L} = v^T \Sigma v - \lambda(v^T v - 1)$, and $\partial \mathcal{L} / \partial v = 0$ yields the eigenvalue problem

$$\Sigma v_i = \lambda_i v_i. \quad (2)$$

The stationary points of the projected variance are the eigenvectors of Σ , and since $v_i^T \Sigma v_i = \lambda_i$, each eigenvalue is the variance along its own direction. Keeping the top k eigenvectors as columns of W_k gives the rank- k subspace of least expected squared reconstruction error (Eckart–Young [4]). Σ is real symmetric, so $W_k^T W_k = I_k$.

2.2 The snapshot trick

Σ is 4096×4096 and singular: it sums N_{train} rank-one terms under one linear constraint, so $\text{rank}(\Sigma) \leq N_{\text{train}} - 1 = 279$ and at most 279 of its 4096 eigenvalues are non-zero. Take instead the Gram matrix $G = \frac{1}{N-1} AA^T \in \mathbb{R}^{N \times N}$. If $Gu_i = \lambda_i u_i$, left-multiplying by A^T gives

$$\frac{1}{N-1} A^T A A^T u_i = \lambda_i A^T u_i \iff \Sigma(A^T u_i) = \lambda_i (A^T u_i), \quad (3)$$

so $A^T u_i$ is an eigenvector of Σ carrying the same eigenvalue. The non-zero spectrum of a 4096×4096 matrix therefore follows from an $N \times N$ eigenproblem, 280×280 when fitting on a training split, and the cost of the decomposition falls from $O(d^3)$ to $O(N^3)$. Benchmarked on the full 400-image matrix this is a reduction of roughly two orders of magnitude. The exact ratio depends on the hardware and on the BLAS implementation, so the notebook measures it on each run rather than fixing a figure here. The resulting vectors are normalised to unit length.

Each v_i lies in $\mathbb{R}^{4096}$, the space of the images themselves, so it reshapes to 64×64 and displays as a *pattern of deviation from the mean face*. Being a linear combination of training faces, $v_i \propto \sum_j u_{ij}(x_j - \mu)$, it inherits their face-like structure, hence the name, and any face is then a recipe of k coefficients over that shared basis: $x \approx \mu + \sum_{j \leq k} z_j v_j$ with $z = W_k^T(x - \mu)$.

2.3 Classification and verification

A test face is projected to $z = W_k^T(x - \mu)$ and classified by k -nearest neighbours [5]. The non-parametric rule is a necessity here rather than a preference. With 40 classes and 7 training images each, a full Gaussian per class would need $k + k(k+1)/2$ parameters from 7 observations. Because W_k is orthonormal, distances in the projected space equal distances between the pixel-space reconstructions. Two variants recur: **plain** projection, where components keep their natural scale so high-variance directions dominate, and **whitened** projection $\tilde{z}_j = z_j / \sqrt{\lambda_j}$, which rescales every component to unit variance and amounts to a Mahalanobis distance inside the retained subspace.

Checked against scikit-learn’s exact SVD solver, the implementation gives a maximum relative eigenvalue error of 5.0×10^{-15} , a minimum absolute eigenvector dot product of 1.0000000000, and a maximum reconstruction difference of 1.6×10^{-14} .

3 Experimental design

Each subject contributes n_{train} images to training and the rest to test, since a uniformly random split would leave some subjects with no training images and so unidentifiable by construction. The default is 7 train / 3 test per subject: $N_{\text{train}} = 280$, $N_{\text{test}} = 120$.

μ and W come from training images only, and test images are projected using those fixed parameters, because fitting PCA on all 400 images before splitting leaks information into the basis and inflates the accuracy reported from it. With 120 test images a single split carries a standard error near 2.4 percentage points, so every figure reported here averages **10** independent splits and carries its standard deviation.

One caveat applies to the choice of k and is stated here rather than left implicit. Two values recur below: $k = 75$, the position of the maximum of the mean accuracy curve, and $k = 25$, the smallest k whose mean accuracy lies within one standard deviation of that maximum. Both were read off the same test splits that the results are reported on, so neither is a clean held-out estimate, and the figures quoted at those k carry some optimism. The accuracy curve bounds it: plain projection moves from 0.926 at $k = 25$ to 0.937 at its peak and stays flat to $k = 279$, a span of about one point, which is within the split-to-split standard deviation of 0.013. Selecting k on a validation partition carved out of the training images would remove the dependence altogether, and is the correct construction where the choice of k has to carry more weight than it does here.

4 Results

Figure 1 shows μ . It is a blurred, generic frontal face, which is what the average of 40 identities under varying expression and lighting should look like: the features that all faces share survive the averaging and everything

Mean face μ (training set)



Figure 1: The mean face μ , averaged over the 280 training images. Every Eigenface below is a pattern of deviation from this image, and every reconstruction begins here.

identifying cancels. Subtracting it is what makes the first component describe variation between faces rather than the shared structure common to all of them.

The first 16 Eigenfaces (percentage = share of total variance)

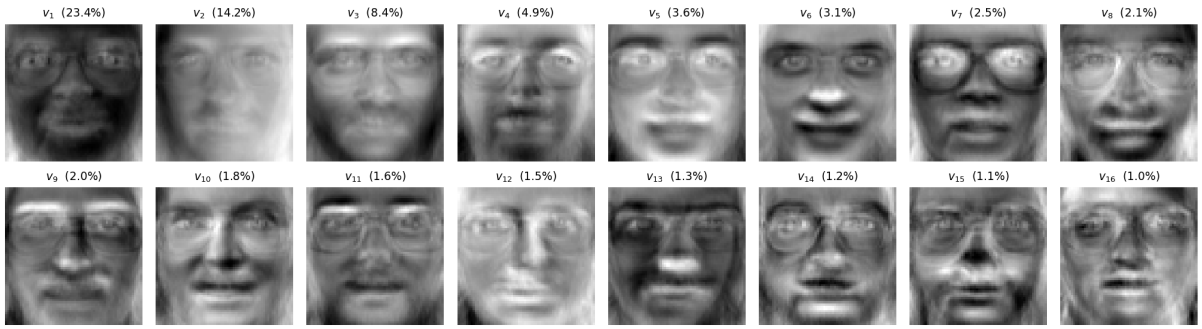


Figure 2: The first 16 Eigenfaces, with each component's share of total variance.

The first three components in Figure 2 are low in spatial frequency and describe whole-image photometric structure rather than facial detail. v_1 is a broad contrast pattern separating the brow and eye band from the rest of the face, v_2 is nearly flat and encodes overall luminance, and v_3 carries a top-to-bottom intensity gradient. None of them describes who the person is. Recognisable structure begins at v_4 : spectacle frames, hairline, moustaches, and changes around the mouth. Later components grow more localised and higher in frequency, carrying identity information alongside sampling noise specific to these training images.

The conventional variance thresholds need 59, 103, and 198 components for 90%, 95%, and 99% respectively, that is 1.44%, 2.51%, and 4.83% of the original 4096 dimensions. The first component alone holds 23.4% of total variance and the first ten hold 66.1%.

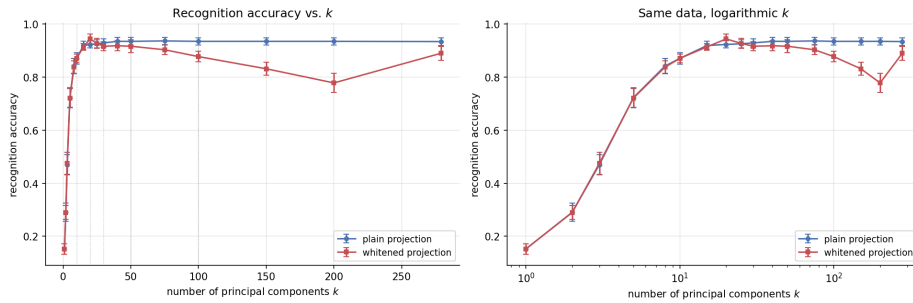


Figure 3: Recognition accuracy against k , plain and whitened projections, mean $\pm$ s.d. over 10 splits.

The two curves in Figure 3 behave differently, and the contrast between them is the clearest result here.

The plain curve rises and then flattens rather than falling, which follows from the geometry rather than from the dataset. W is orthonormal, so with every component retained the Euclidean distance in Eigenface space equals the pixel distance exactly, and adding components moves 1-NN monotonically towards the full-dimensional answer. Trailing components carry tiny eigenvalues and contribute almost nothing to a Euclidean distance, so under this metric keeping too many components costs computation rather than accuracy.

Table 1: Recognition accuracy against the number of retained components.

k	Plain	Whitened	k	Plain	Whitened
10	0.871 ± 0.021	0.871 ± 0.017	50	0.935 ± 0.015	0.917 ± 0.025
20	0.923 ± 0.014	0.944 ± 0.019	75	0.937 ± 0.013	0.903 ± 0.018
25	0.926 ± 0.016	0.928 ± 0.020	100	0.935 ± 0.013	0.878 ± 0.020
30	0.929 ± 0.016	0.917 ± 0.017	200	0.935 ± 0.013	0.778 ± 0.036

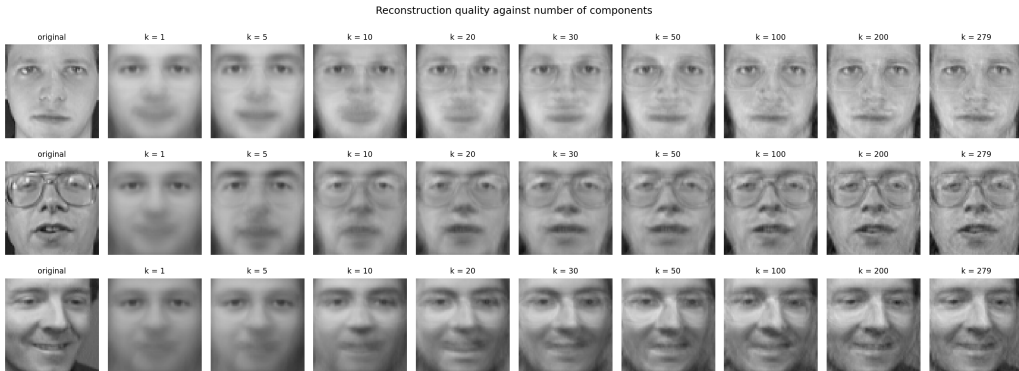
The whitened curve rises, peaks at $k = 20$, then falls from 0.944 to 0.778 at $k = 200$. Whitening divides component j by $\sqrt{\lambda_j}$, which for a trailing component means dividing by a very small number and so promoting it to equal footing with the leading ones. The i -th eigenvalue, however, comes from 280 observations in 4096 dimensions. For large i that estimate is dominated by sampling noise, and the direction then describes quirks of the particular training images rather than a property of the population. Whitening amplifies exactly that noise, producing the familiar overfitting signature.

Both curves point at one principle. The useful number of components is set by how many directions the sample estimates reliably, not by how many exist mathematically, and whether the unreliable ones do damage depends on whether the metric gives them weight. That is why the curves diverge, and why “how many components should I keep?” has no answer the representation alone can supply.

At $k = 75$, 1-NN with Euclidean distance gives accuracy 0.9367 ± 0.0137 , macro precision 0.9507, macro recall 0.9367, and macro F_1 0.9339. Macro-averaging weights every subject equally, so failure on a few individuals cannot hide behind a good average. On the reference split 8 of 120 test images are misclassified and the confusion matrix is almost entirely diagonal, with no pair of subjects mutually confused.

Across distances at $k = 75$, Manhattan reaches 0.9442 against Euclidean’s 0.9367 and cosine’s 0.9300 at 1-NN, a margin well under one standard deviation. The dominant effect is neighbourhood size rather than the distance function. Euclidean accuracy falls to 0.854 at 3-NN and 0.805 at 5-NN. With 7 training images per subject a 5-NN vote often reaches past the correct identity into neighbouring clusters, so the usual argument that larger n smooths noise assumes a sample density this problem lacks.

4.1 Reconstruction

**Figure 4:** Reconstructions of three test faces at increasing k .

Reconstruction maps a projection back by $\hat{x} = \mu + W_k W_k^T (x - \mu)$, and since $W_k W_k^T$ is an orthogonal projector, the residual lies wholly in the discarded directions, giving $\mathbb{E}\|x - \hat{x}\|^2 = \sum_{j>k} \lambda_j$. In Figure 4, at $k = 1$ every face is the mean face with a photometric adjustment. Identity becomes visually recognisable around $k = 20$ to 30, and past $k \approx 100$ the changes are confined to fine texture. Test RMSE falls from 0.083 at $k = 10$ to 0.050 at $k = 100$ and 0.041 at $k = 279$, with PSNR 21.8, 26.2, and 28.1 dB.

Reconstruction error keeps falling well past the point where recognition stops improving, because the variance removed across that range, fine skin texture, sensor noise, small expression detail, feeds pixel fidelity and not identity. Holding 95% of variance costs about four times the components recognition needs, $k = 103$ against $k = 25$, and buys roughly one accuracy point, which sits inside the split-to-split noise. A variance threshold is therefore the wrong criterion for choosing k in a recogniser: it measures how well the subspace reproduces images, where recognition needs it to separate identities.

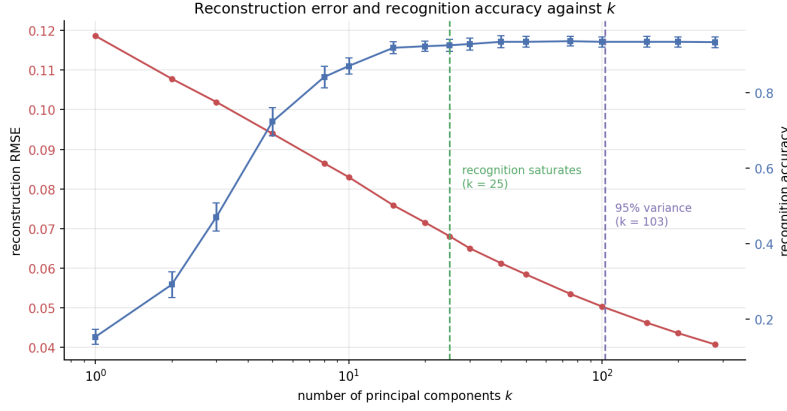


Figure 5: Reconstruction error and recognition accuracy against k .

5 Critical analysis

Enrolment set size is the strongest single influence measured. Accuracy runs 0.547, 0.698, 0.774, 0.820, 0.867, 0.923, and 0.950 for 1, 2, 3, 4, 5, 7, and 9 training images per subject, and has not saturated at nine. Two mechanisms compound. The classifier holds few reference points, so a test image with an unseen expression has no near neighbour of the correct identity; and the basis itself degrades, since at $n_{tr} = 1$ only 39 components exist, estimated from 40 images. The 40-point drop from nine images to one is the familiar one-sample-per-person difficulty.

Illumination. Test images were multiplied by a left-to-right linear intensity ramp of increasing strength, modelling a directional source absent at enrolment, with training images untouched.

Table 2: Recognition accuracy under increasing illumination mismatch.

Ramp strength	0.0	0.2	0.3	0.4	0.5	0.7
Euclidean	0.937	0.878	0.758	0.488	0.305	0.142
Cosine	0.930	0.869	0.726	0.471	0.308	0.181
Drop first 3	0.932	0.938	0.936	0.920	0.890	0.783

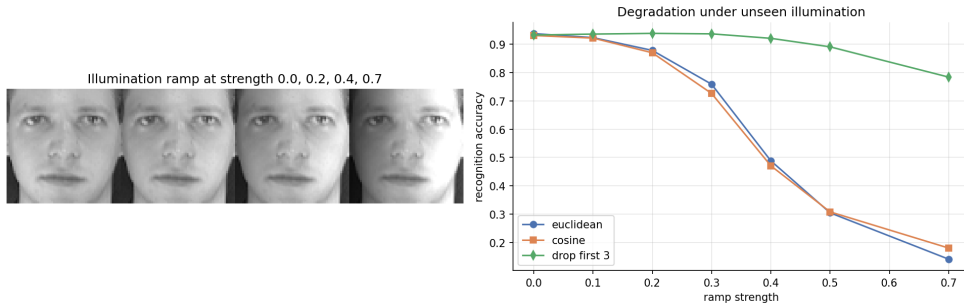


Figure 6: Left: the ramp at strengths 0.0, 0.2, 0.4, 0.7. Right: accuracy under increasing illumination mismatch.

Euclidean 1-NN collapses. Of the two mitigations tested, only one works.

Cosine distance does not help, tracking the Euclidean result at every strength, because the expectation that it would help assumed a uniform brightness change, which shifts a projection mainly in magnitude. This ramp varies spatially, brightening one side of the face and darkening the other, so it rotates the projection rather than lengthening it, and discarding magnitude discards nothing relevant. Invariance to global scaling buys little against lighting with spatial structure, which real lighting always has.

Discarding the first three components does work. At the strongest ramp Euclidean 1-NN falls to 0.142 while the same classifier without those components holds 0.783, so those three directions absorb nearly the whole perturbation. This is experimental support for the reading of the component structure given earlier: the leading eigenfaces encode photometric conditions rather than identity. Discarding the three directions that carry the largest share of variance in the entire dataset improves recognition under illumination change. At the top of the spectrum, variance ranking and discriminative value work against each other.

Under matched illumination, dropping up to three components moves accuracy by less than one standard deviation either way, and dropping beyond about five costs real accuracy. Removing three discards 46% of total variance and leaves recognition unchanged. Discarding leading components is therefore insurance against mismatched conditions rather than a free gain.

6 Answering the research question

1-NN on the raw 4096 pixels, with no reduction, gives 0.9342 ± 0.0142 .

Table 3: Dimensionality that can be removed at three tolerance levels.

Criterion	k	Accuracy	Dimensions removed	Variance retained
Matches peak	75	0.937	98.17%	92.4%
Within 1% of peak	30	0.929	99.27%	82.7%
Within 5% of peak	15	0.919	99.63%	72.7%

About **99% of the original dimensions can be discarded with no measurable loss of identity information**. At $k = 25$ the Eigenface representation scores 0.9258 ± 0.0160 against the raw-pixel 0.9342 ± 0.0142 , nominally lower by well under the split-to-split noise, so the two are indistinguishable at this sample size. The 164-fold compression is free rather than beneficial.

The computational claim needs the same care. Measured end to end the Eigenface pipeline runs slower than raw 1-NN, 17.8 ms against 3.5 ms, because it includes fitting PCA on every split. The reduction repays itself at query time and at scale, since each comparison costs 25 multiplications instead of 4096 and the gallery shrinks by the same factor, but across 400 images that saving does not cover the one-off decomposition.

Three qualifications hold the headline in place. The bound comes from sample size, not from faces: the training set admits at most $N_{\text{train}} - 1 = 279$ components, so achievable compression depends on how many images were available rather than on any intrinsic dimensionality of human faces. The threshold depends on the criterion, since under reconstruction the answer is several times larger. It depends on operating conditions too, because every figure here comes from images captured much like the training images. A compression lossless under matched conditions need not stay lossless under mismatched ones.

7 Limitations

ORL faces are cropped, frontal, aligned, and captured against a constant background in one laboratory, with no clutter, occlusion, or age variation. No detection or alignment stage was required. In deployed systems alignment error is often the dominant failure source, so these accuracies do not estimate unconstrained performance.

PCA is unsupervised and maximises total variance without reference to labels, which is why the leading components encode illumination. Fisherfaces [3], maximising between-class to within-class scatter, typically does better under lighting variation. That comparison was not implemented here, however, so the claim that PCA is suboptimal rests on the observed component structure and the illumination experiment rather than on a direct measurement.

The model is linear, while pose and expression move a face along a curved manifold a linear subspace only approximates. The scale is small. Even across ten splits the standard deviation at the operating point is about 1.5 percentage points, so differences of a point or two carry no interpretation here. Accuracy also falls as the gallery grows, so these figures would not transfer to thousands of identities. The illumination model is synthetic. It clips at unity, so saturation is represented, and at the strongest ramps a substantial part of the brightened side of each image is clipped; but it omits cast shadows and specularities, both non-linear effects likely to degrade performance further. The setting is closed-set: nearest-neighbour classification always returns some identity, whereas a deployed system needs a rejection threshold, a separate problem.

8 Conclusions

The snapshot trick recovers the eigenfaces from an $N \times N$ eigenproblem instead of a 4096×4096 one, roughly two orders of magnitude faster, matching a reference implementation to machine precision. The leading eigenfaces

encode photometric conditions rather than identity, a consequence of ranking directions by variance and confirmed by the illumination experiment. Recognition accuracy saturates at a few tens of components while reconstruction error keeps falling, so the conventional 95% variance threshold calls for roughly four times the components recognition needs. Retaining more components does not necessarily generalise better, and whether it hurts depends on the metric. Plain Euclidean 1-NN plateaus, where a whitened metric degrades as noise-dominated components are amplified. Enrolment set size dominates every other factor tested, costing about 40 percentage points between nine training images per subject and one. Unseen illumination is the most damaging perturbation, and the effective remedy is to discard leading components rather than to change the distance measure. Cosine distance, often recommended for this purpose, gave no benefit against spatially varying lighting.

All results come from `notebooks/eigenfaces_project.ipynb`, which runs top to bottom, writes every figure to `figures/` and every number to `report/results.json`, and uses fixed seeds so the reported values reproduce exactly.

References

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